DFE pharma

OPTIMISE YOUR **DC FORMULATION**

Dealing with Direct Compression formulation challenges



Dealing with Direct Compression formulation challenges

Direct compression requires the use of particle engineered excipients with the right characteristics for your specific formulation. This booklet intends to help formulators interested in DC in two parts:

- Part 1 provides an overview of potential formulation challenges and ways to overcome them.
- Part 2 showcases the DC grade portfolio of DFE Pharma, and provides recommendations on specific characteristics.

Please keep in mind that the optimal formulation and choice of excipients will depend on the specific situation, and therefore we always recommend discussion with our technical experts. Feel free to reach us for advice via your regional contact, or via 'Ask and Expert' tab on our website (dfepharma.com).

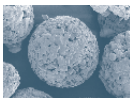
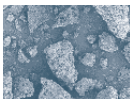
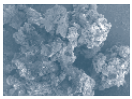
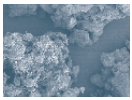
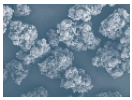
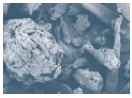
Your excipient solution provider



Part 1: Direct compression challenges: an overview of potential formulation challenges and ways to overcome them.

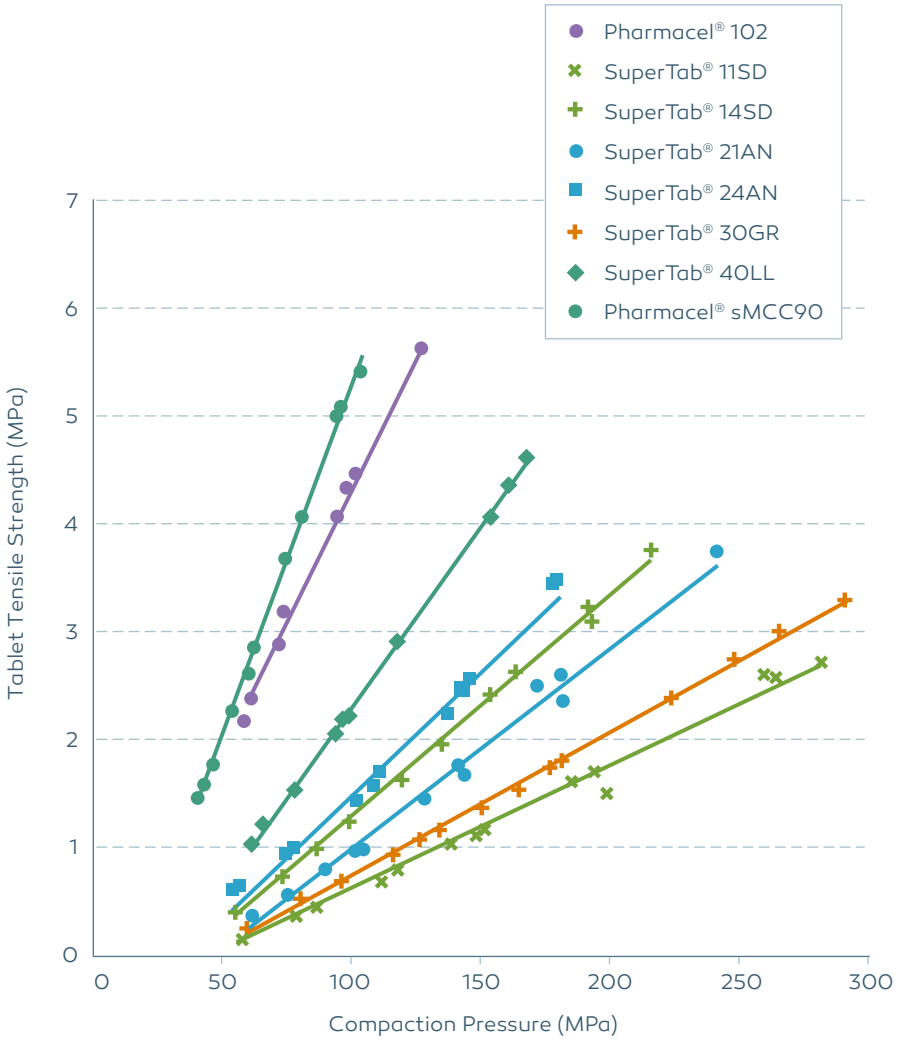
All DC grades are developed to provide good compaction and good flow. The table indicates functional differentiators.

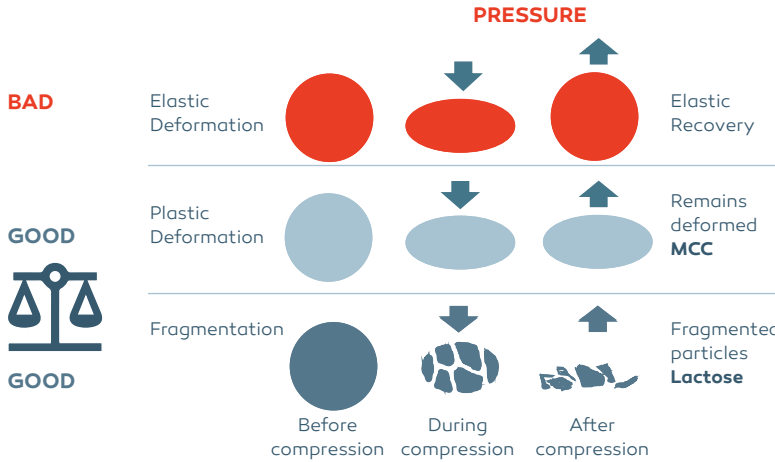
■ Baseline
\+ Good
++ Very good
+++ Superior

	Compaction	Re-compaction	Flowability	Disintegration	Content Uniformity
 **Spray dried lactose** SuperTab® SD Lactopress® SD	■(+)[2]	■	+++	+	+
 **Anhydrous lactose** SuperTab® AN	+	+++	+(++)[1]	■	+
 **Granulated lactose** SuperTab® 30GR	■	■	++	+	++
 **Granulated anhydrous lactose** SuperTab® 24AN	++	+	++	■	++
 **Co-processed** SuperTab® 40LL Pharmacel® sMCC90	+++	■	+++	■	++
 **MCC** Pharmacel®	+++	■	■	++	■
 **Superdisintegrant** Primojel® Primellose®	■	■	■	+++	■

[1] *SuperTab® 22AN with +++ flow available*
[2] *SuperTab® 14SD with + compactability available*

Tablet tensile strength and friability can be improved by adjustment of the formulation or process parameters





Physico-mechanical properties of powders determine the compaction behaviour

Improve compaction by adjustment of the physical properties of the formulation

- Select excipients with highest compaction
- Use SuperTab® 40LL as most compactible soluble filler
- Use SuperTab® 24AN as most compactible lactose filler
- Increase the amount of filler in the formulation, by increasing tablet weight and size
- Add or increase the amount of tablet binder in the formulation
- Increase Pharmacel® content of a lactose based formulation
- Reduce lubrication amount for lubricant sensitive formulations

Improve compaction by adjustment of process parameters

- Reduce lubrication blending intensity for lubricant sensitive formulations
- Increase compaction force
- Optimise pre-compaction
- Increase dwell time by reducing tableting speed
- Change tablet shape

The recommended grade for dry granulation is SuperTab® 21AN, as it exhibits little or no loss of tabletability upon re-compaction.

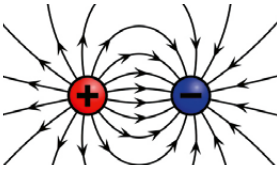
Flowability can be improved by adjustment of physical properties or by changing the flow field or stress state



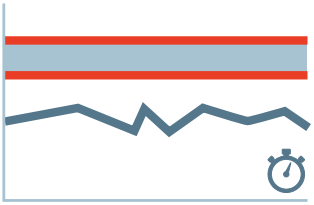
1
Arching



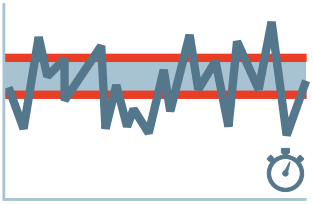
2
Ratholing/
funneling



3
Electrostatics



4
Mass flow limitation



5
Inconsistent flow

"Flowability is the result of material physical properties and the equipment used for handling, storing, or processing material. Flow behaviour (and ranking) of powders may be different between applications, as it depends on the stress state and flow field."

Improve flow by adjustment of the physical properties of the formulation

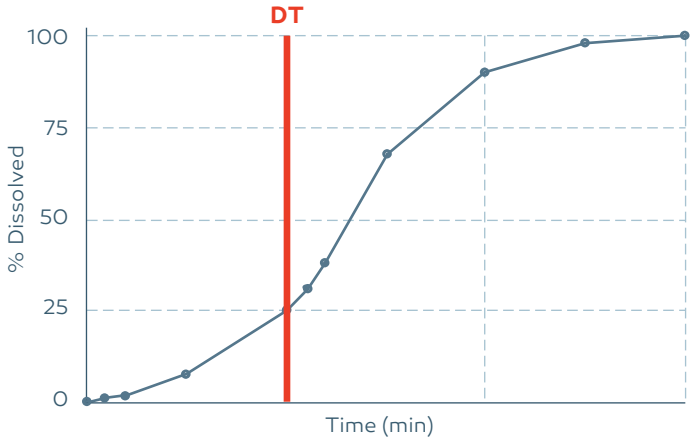
- Reduce cohesiveness, internal friction (1,5)
- Reduce wall friction (2,5)
- Reduce electrostatics by using low charging excipients (3)
- Increase density (4)
- Increase permeability (4)
- Increase homogeneity of the formulation (5)
- Increase ratio DC lactose: MCC
- Add flow aids/glidants
- Optimise PSD: coarse particles are generally less cohesive
- If dissolution allows, coarsen API
- Optimise shape and surface: spherical particles (spray dried/granulated lactose) tend to have better flow properties

Improve flow by adjustment of the flow field or stress state

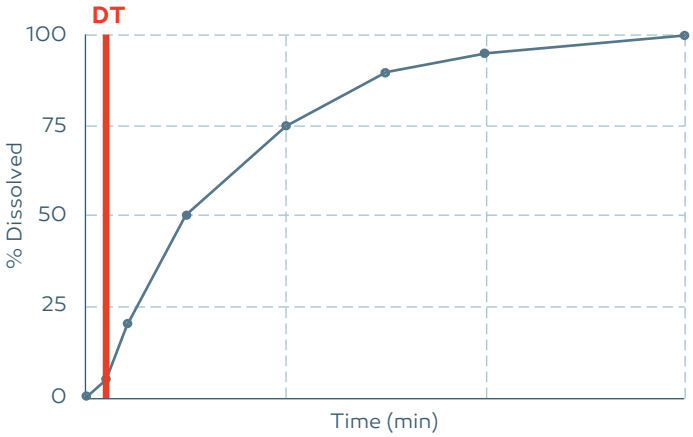
- Increase hopper/feeder aperture (1, 4, 5)
- Increase time for the formulation to flow into the dies by slowing down the tablet press (4,5)
- Reduce hopper wall angle (2, 5)
- Reduce electrostatics by reducing frictional contacts, grounding of the feeder, coating or electro-polishing equipment (3)
- Reduce hopper fill height
- Aerate powders by mechanical assistance (paddle feeders, blades, agitators)
- Optimise humidity to reduce electrostatics and capillary interactions

(*) numbers indicate prefered way forward for specific issues

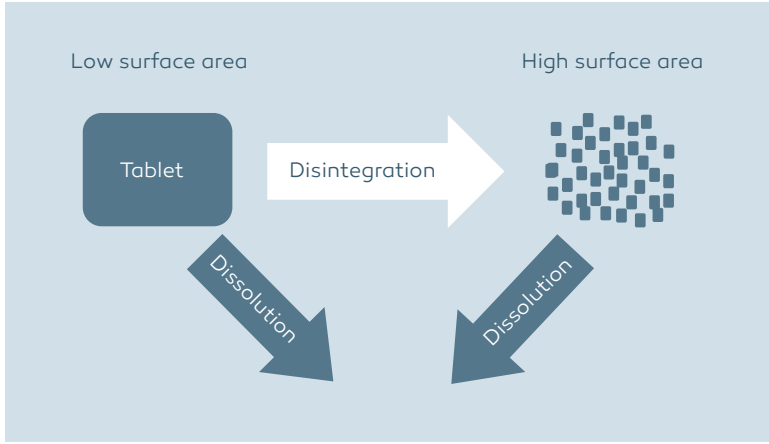
Dissolution can be improved by improved disintegration (when this is limiting) or by adjustment of API properties



1. Dissolution curve is limited by the disintegration



2. Dissolution curve is not limited by the disintegration; dissolution is limited by the solubility of the API



Disintegration increases the available surface area to increase drug availability for dissolution and subsequently absorption

Improve dissolution by reduction of the disintegration time, generally driven by the disintegrant

- Increase amount of disintegrant
- Add Pharmacel® to aid disintegration of lactose based formulations
- Optimise type of disintegrant
- SuperTab® 30GR is the quickest disintegrating DC lactose grade
- Decrease compaction to reduce hardness

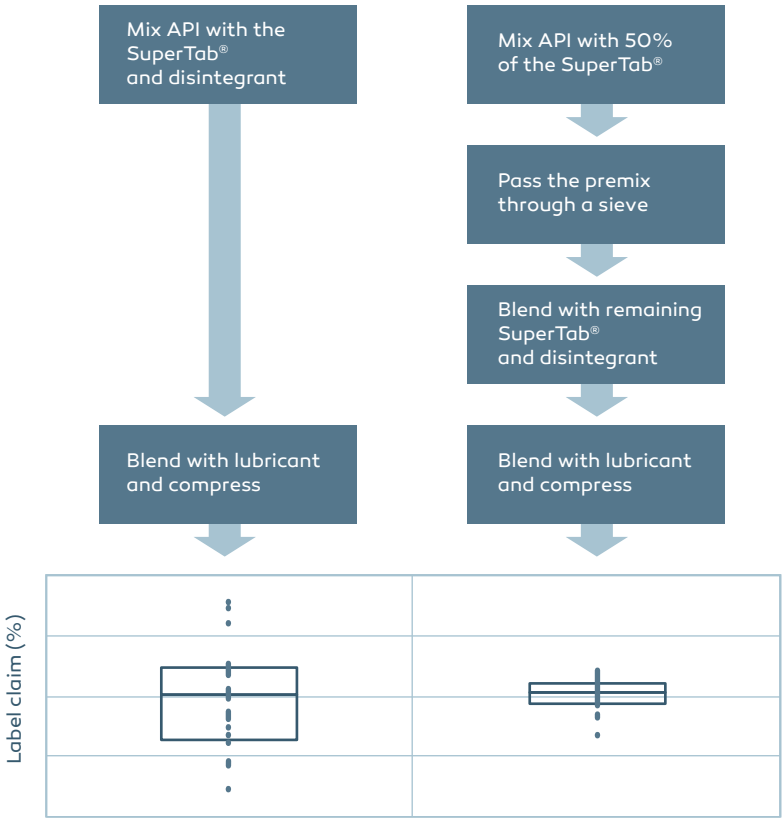
Improve dissolution by adjustment of the API properties to improve solubility of API

- Increase surface of API
- Modify surface properties of API
- Add solubilizer
- In some cases it might help to intimately mix API with excipient with high solubility and high adhesion to API to aid dissolution
- Make solid dispersions with Primojel®

Content uniformity can be improved by effective dispersion of the API

	Low Dose	Medium Dose	High Dose
% dosage	< 2%	2 - 20%	> 20%
CU concerns	Primary*	Minor	Unlikely >25%

** Even at very low dose (0.005%w/w), good content uniformity may be achieved when a proper mixing plan is used.*

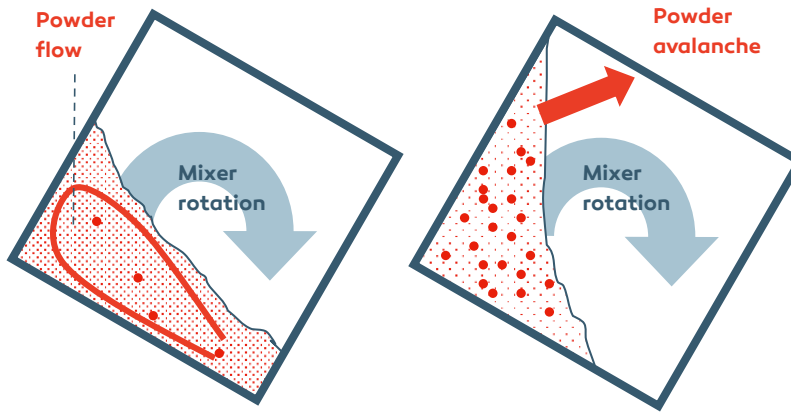


Improve content uniformity by employment of an effective mixing plan to disperse cohesive agglomerates

- Use layering during addition of API
- Pre-blend API with part of (good flowing) excipients and de-agglomerate (e.g. sieve) before blending in other excipients
- Increase shear during blending

Improve content uniformity by adjustment of excipient or API properties

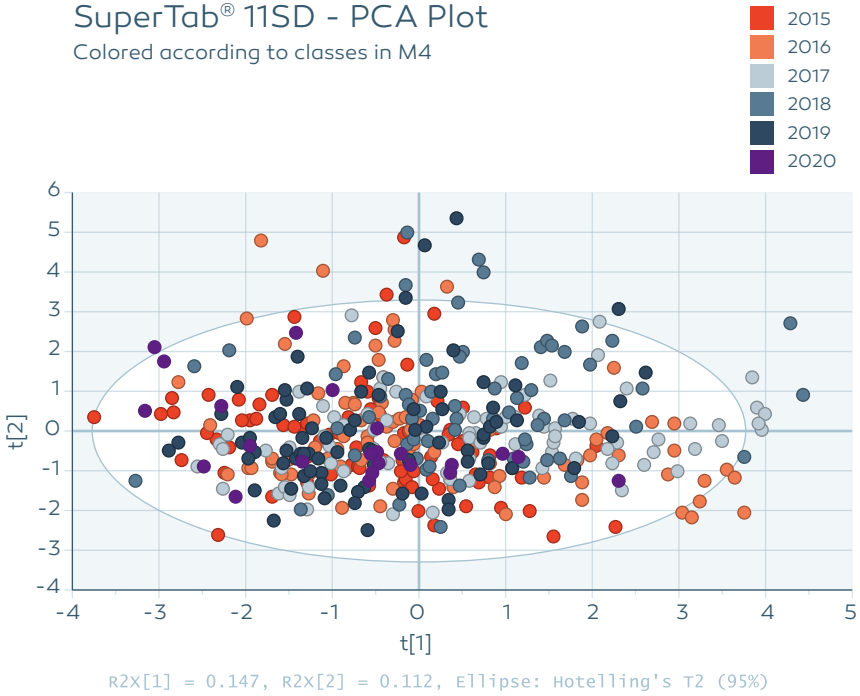
- Select filler-binders with good flow properties and high surface area that help to disperse agglomerates during mixing, like granulated and spray dried lactose
- Reducing the API size results in more particles to disperse
- Segregation can be minimised by the structure of granulated lactose
- Segregation can be minimised by adding a milled or sieved lactose grade with similar particle size as the API
- Segregation can be minimised by use of excipients which have good adherence to the API



Free flowing powder tends to set up a 'rolling' motion inside the powder bed, which contributes to de-agglomeration of the API. Poorly flowing powder tends to slump or avalanche, which is less effective for de-agglomeration.

Robustness of a formulation can be improved by using ingredients that are consistent and have low process sensitivity

"Principle Component Analysis (PCA) is a multivariate technique used by DFE Pharma to investigate patterns and clusters in large product data sets. A high degree of overlap of datapoints shows that there is a high batch-to-batch and year-to-year consistency."



Principal Component Analysis score plot of SuperTab® 11SD batches produced between 2015 and 2020. All analyzed batches are well within specification and the ellipse indicates the 95 percent Hotelling's interval. The combined R2X of the two components is 0.26, indicating little trend in the data. High overlap between data points shows that there is a high degree of consistency between these batches.

Improve robustness of a formulation by proven consistency

- Understanding excipient variability can de-risk the use of excipients by ensuring the use of the same physical and chemical properties during development and production
- Ensured year-after-year, batch-after-batch consistency provides confidence in product consistency over the entire marketed period of a drug product

Improve formulation robustness by the use of excipients with low process sensitivity

- Switch to less elastic excipients to reduce dwell time and lubricant sensitivity (e.g. switch from MCC to lactose)
- Increase surface area of excipients to reduce lubricant sensitivity
- Select excipients with optimal morphology to reduce lubricant sensitivity; flat surfaces delaminate lubricant more easily and are therefore more sensitive to over lubrication
- Granulated lactose grades provide the lowest lubricant sensitivity
- SuperTab® 40LL provides the lowest dwell time sensitivity
- Plastic excipients, like MCC, often result in low ejection forces which can increase the tooling lifespan



DFE pharma

Product Consistency
Batches released in 2019

Product group: Lactose
Brand name: SuperTab® 30GR
Article code: 743715 / 743716
Product description: Granulated lactose
Document No.: PD-0626 Page 1 of 3

Dear Customer,

The following is a summary of selected batch data showing the consistency of our product, SuperTab® 30GR, over the year 2019. Data shown is for the identified potential functional related characteristics (FRCs, termed CMAs or critical material attributes by the U.S. Pharmacopoeia).

To the best of our expert knowledge and referencing the European Pharmacopoeia monograph for lactose (lactose monohydrate) potential FRCs identified are:

- The particle size distribution – indicator of powder flow and the segregation potential
- Bulk and tapped density – indicator of powder flow, mixing (fill volume) and tablet die filling
- Water content as measured by the Karl Fischer method – indicator of total moisture

The graphs below show for each FRC the values measured by QC (Quality Control) for 11 (eleven) batches released in the year 2019. Dashed lines represent product specification limits. For the particle size measurements an airjet sieve is used. The bulk- and tapped density are measured conform USP-616> Type I.

Particle size (% w/w)

Density (kg/m³)

< 25 micron
< 355 micron
> 450 micron
Bulk
Tapped

DFE pharma

Product Consistency
Batches released in 2019

Product group: Lactose
Brand name: SuperTab® 30GR
Article code: 743715 / 743716
Product description: Granulated lactose
Document No.: PD-0626 Page 2 of 3

Moisture (Karl Fischer) (% w/w)

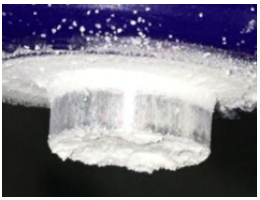
All the potential functional related characteristics show a consistent measurement value.

The table below lists the measurement values found in the graphs. The batch order is the same as in the graphs above.

Batch	PSD < 25 µm (% w/w)	PSD <250 µm (% w/w)	PSD <355 µm (% w/w)	Bulk Density (kg/m³)	Tap Density (kg/m³)	Karl Fischer (% w/w)
1	19	97	98	556	688	4.6
2	18	98	97	575	719	4.7
3	20	97	96	548	679	4.7
4	16	94	96	569	697	4.7
5	16	96	96	540	676	4.6
6	19	97	97	557	676	4.7
7	15	94	96	538	672	4.6
8	20	97	96	580	704	4.6
9	18	98	97	585	693	4.6
10	17	95	96	599	666	4.7
11	19	99	97	569	683	4.6

Punch sticking can be reduced by increase of the excipient-API interaction or by decrease of the API-punch interaction

"Punch sticking is the adherence of powder to the tooling surface."





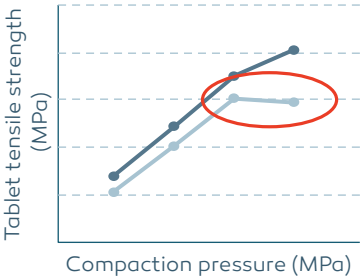
$$\text{Sticking ratio} = \frac{\text{Tablet tensile strength}}{\text{Adhesion}} = \frac{\frac{2 \cdot P}{\pi \cdot d \cdot t}}{\left(\frac{\text{Punch adhesion}}{\text{Punch area}}\right)}$$

*P=tablet hardness, d=diameter, t=thickness

Capping or lamination can be reduced by minimising elastic deformation, via adjustment of the process or formulation

"Capping or Lamination is the horizontal separation of a tablet due to air entrapment in the formulation"





Reduce punch sticking by reduction of the adherence of the API to the punch

- Use excipients with higher tabletability to increase the API-excipient interaction
- Increase efficiency of lubrication (type, amount, mixing)
- Add flow-aid or anti-adherent (e.g. silica, talc)
- Reduce API surface area by increasing PSD, if dissolution allows
- Optimise punch design; e.g. reduce embossing, polish/coat punches, different type of steel
- Optimise pre-compression and compression pressure
- Increase dwell time
- Reduce humidity and moisture in formulation
- Reduce heat of compression for low temperature melting material (API)
- Use filler-binders with lower surface area, lower electrostatics and lower hygroscopicity (e.g. lactose generally has lower punch sticking propensity than mannitol)

Reduce capping/lamination by minimising the elastic deformation relative to the tensile strength

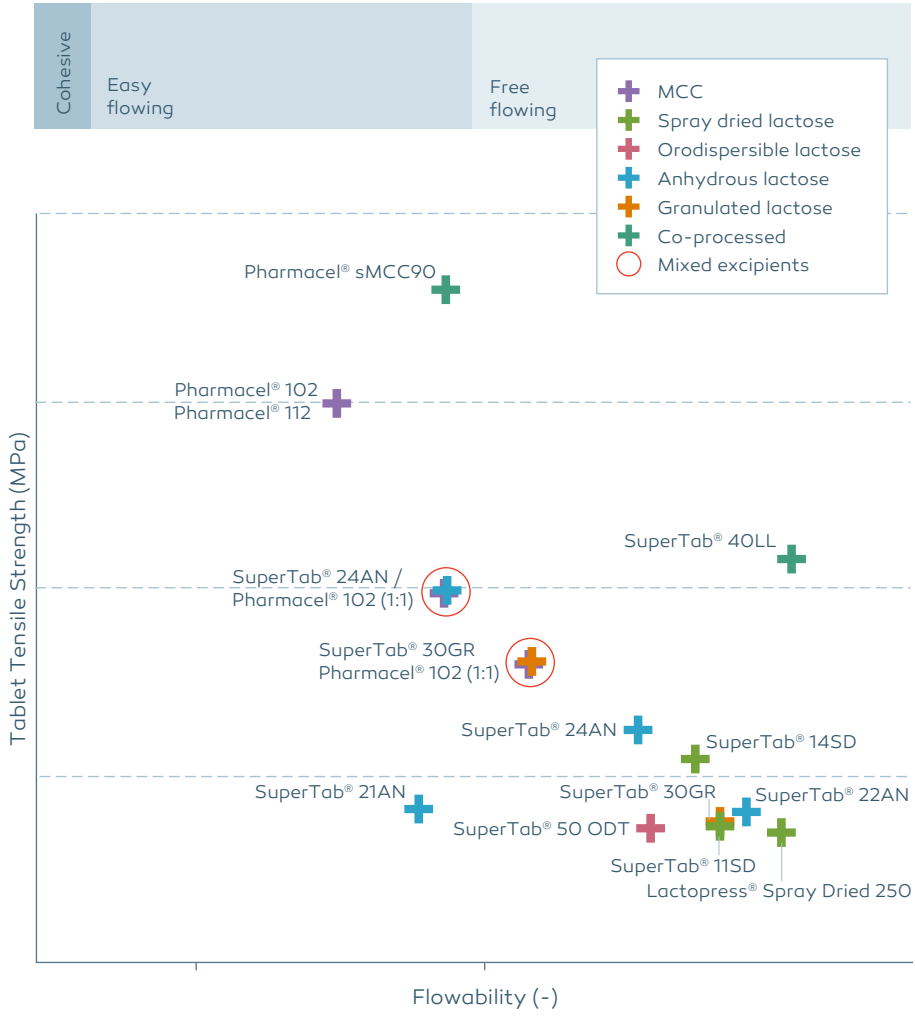
- Optimise formulation to reduce elastic recovery (e.g. reduce starch content)
- Reduce the amount of fines in the formulation
- Use proper tooling; damaged punches promote capping and worn die cavities can result in slip-stick behaviour
- Reduce air entrapment by optimisation of pre-compression
- Reduce air entrapment by using tapered dies
- Reduce the tablet thickness by different shaped tablets (e.g. change from deep concave punches to flat beveled)
- Decrease compression pressure
- Increase dwell time
- Optimise lubrication

Overview typical values physical chemical properties

Product*	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Superdisintegrants						
• Primojel®	4.8	0.79	0.96	20	40	70
• Primellose®	5.3	0.55	0.74	20	50	110
MCC						
• Pharmacel® 102	3.8	0.31	0.44	40	90	180
• Pharmacel® 112	1.3	0.33	0.46	40	90	180
Spray dried lactose						
• Lactopress® Spray Dried 250	0.2	0.64	0.74	45	110	200
• SuperTab® 11SD EU	0.3	0.60	0.72	50	120	220
• SuperTab® 11SD NZ	0.2	0.62	0.69	50	140	240
• SuperTab® 14SD	0.3	0.62	0.72	50	130	240
Orodispersible lactose						
• SuperTab® 50ODT	0.2	0.64	0.73	55	130	220
Anhydrous lactose						
• SuperTab® 21AN	0.1	0.72	0.91	15	170	340
• SuperTab® 22AN	0.1	0.68	0.80	70	220	380
• SuperTab® 24AN	0.1	0.49	0.60	40	140	300
Granulated lactose						
• SuperTab® 30GR	0.1	0.55	0.67	40	140	300
Co-processed						
• SuperTab® 40LL	0.2	0.54	0.66	80	180	350
• Pharmacel® sMCC90	3.8	0.31	0.47	35	110	270

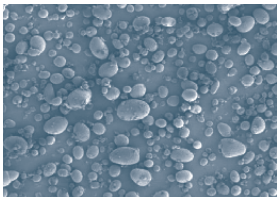
** See www.dfepharma.com for the full portfolio*

Part 2: Direct compression solutions: an overview of the DC portfolio of DFE Pharma with recommendation on specific characteristics.



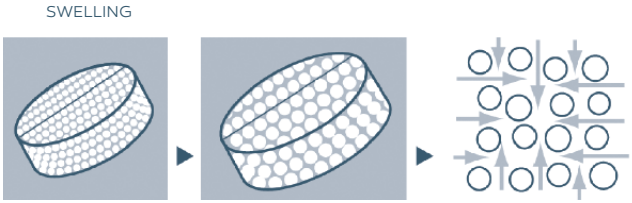
** Flow evaluation based on Schulze shear cell measurements*

Superdisintegrants



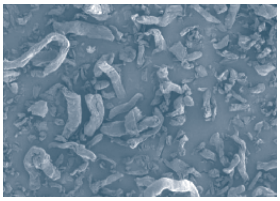
Primojel® (Sodium Starch Glycolate)

- Disintegration by powerful swelling
- Absorbs up to 20 times its own weight
- Traditionally recommended for insoluble matrices



Primellose® (Croscarmellose sodium)

- Disintegration by rapid wicking
- Traditionally recommended for soluble matrices



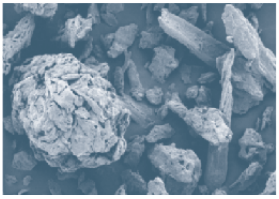


Water is pulled into pores by disintegrant and reduce the physical bonding forces between particles

Product	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Superdisintegrants						
• Primojel®	4.8	0.79	0.96	20	40	70
• Primellose®	5.3	0.55	0.74	20	50	110

** Typical values*

Microcrystalline Cellulose



Pharmacel®

- Highest compactibility

Low moisture

Pharmacel® 102 | **Pharmacel® 112**

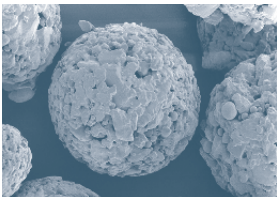


Maximising supply security with two production sites

Product	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
MCC						
• Pharmacel® 102	3.8	0.31	0.44	40	90	180
• Pharmacel® 112	1.3	0.33	0.46	40	90	180

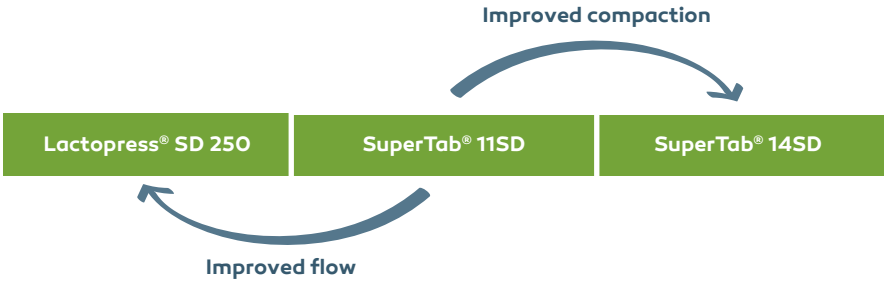
** Typical values*

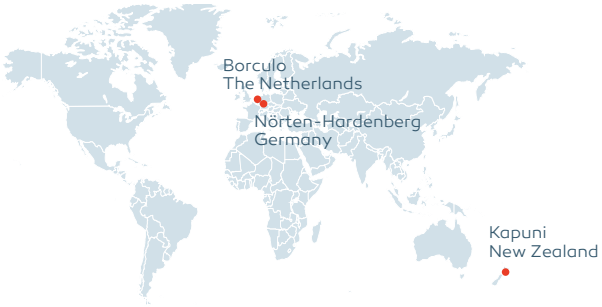
Spray dried lactose



SuperTab® and Lactopress®

- Free flowing
- Excellent dispersibility
- Dual sourced



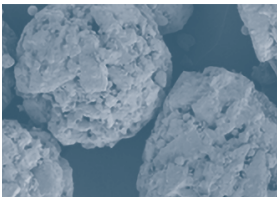


A global offering supplied from multiple locations

Product*	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Spray dried lactose						
• Lactopress® Spray Dried 250	0.2	0.64	0.74	45	110	200
• SuperTab® 11SD EU	0.3	0.60	0.72	50	120	220
• SuperTab® 11SD NZ	0.2	0.62	0.69	50	140	240
• SuperTab® 14SD	0.3	0.62	0.72	50	130	240

** Typical values*

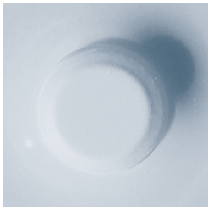
Orodispersible lactose



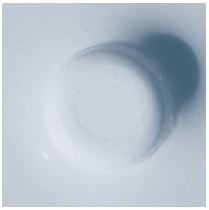
SuperTab® 50 ODT

- Fast disintegration
- Pleasant mouthfeel
- Low friability
- Low sticking propensity
- 100% natural, soluble material

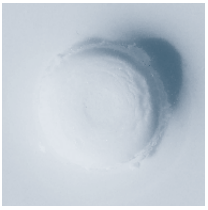
SuperTab® 50 ODT



0 seconds



30 seconds

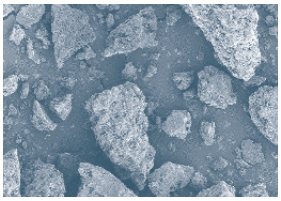


60 seconds

Product*	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Orodispersible lactose						
• SuperTab® 50ODT	0.2	0.64	0.73	55	130	220

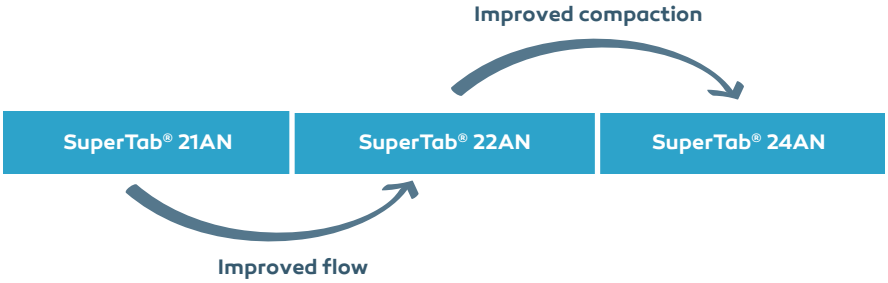
** Typical values*

Anhydrous lactose



SuperTab®

- Low moisture content
- High compactibility
- Excellent re-compactibility



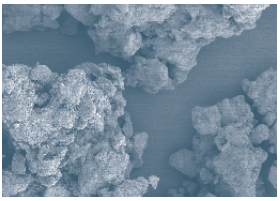


Maximising supply security with two parallel lines

Product*	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Anhydrous lactose						
• SuperTab® 21AN	0.1	0.72	0.91	15	170	340
• SuperTab® 22AN	0.1	0.68	0.80	70	220	380
• SuperTab® 24AN	0.1	0.49	0.60	40	140	300

** Typical values*

Agglomerated Anhydrous lactose



SuperTab® 24AN

- Excellent flow
- Quick disintegration time
- Superior compaction
- Minimum moisture content
- Low lubricant sensitivity

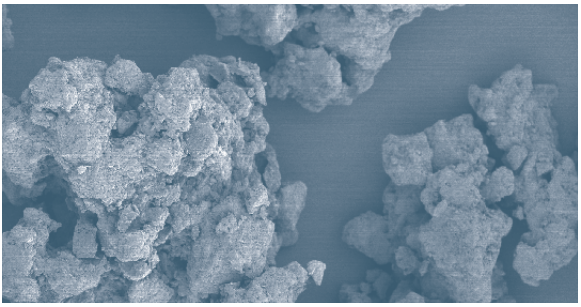
SuperTab® 24AN: Combining the best of both worlds

Agglomerated

- High (powder) flowability
- Excellent mixing properties
- Quick tablet distintegration
- High compactability

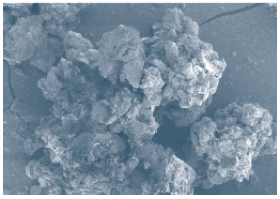
Roller dried

- Low moisture content
- High compactibility



SuperTab® 24AN outperforms other marketed anhydrous grades in both flow and compactibility with its unique morphology, generated by an additional granulation process.

Granulated lactose



SuperTab® 30GR

- Good flow and compaction
- Low segregation potential
- High formulation robustness

SuperTab® 30GR

SuperTab® 30GR:

Supports the process of formulation development



Low dwell time sensitivity



Low moisture sensitivity

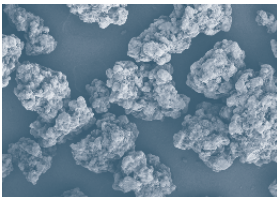


Low lubricant sensitivity

Product*	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Granulated lactose						
• SuperTab® 30GR	0.1	0.55	0.67	40	140	300

** Typical values*
Other advanced functionality granulated products are displayed in other tabs (SuperTab® 24AN: Anhydrous, SuperTab® 40LL: Co-processed)

Co-processed

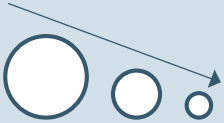


SuperTab® 40LL (Lactose-lactitol)

- Excellent flow and compaction
- Excellent formulation robustness
- Fully soluble

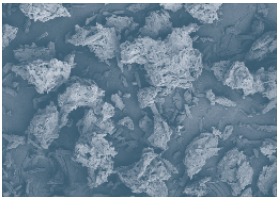


Reduce the size of your tablets



Pharmacel® sMCC90 (Silicified MCC)

- Excellent compaction and flow
- Improved formulation robustness compared to MCC
- Monograph available



Co-processed excipients redefine the limits of direct compression for high dose formulations

Product*	Loss on Drying (%w/w)	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Co-processed						
• SuperTab® 40LL	0.2	0.54	0.66	80	180	350
• Pharmacel® sMCC90	3.8	0.31	0.47	35	110	270

** Typical values*

Global Presence

1 Formulation support

2 Ask an expert

3 Product information

4 Sample request

5 Digital tools

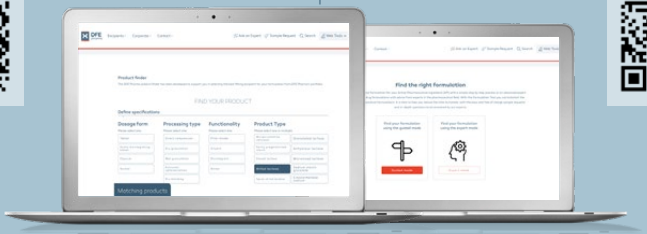
6 Technical documents

Visit our website!

dfepharma.com

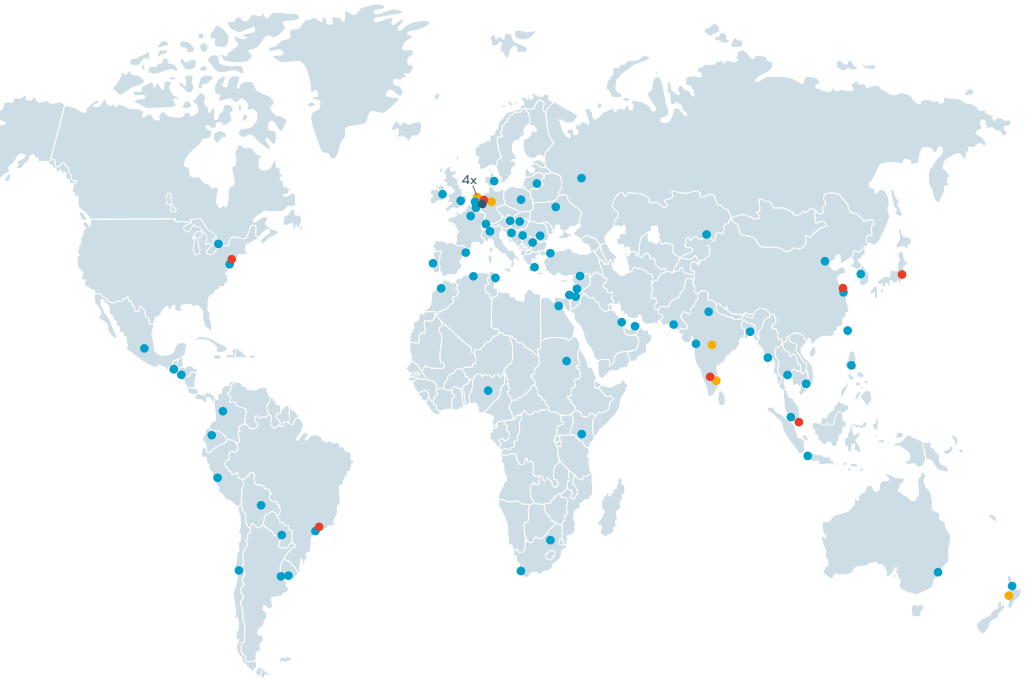
Product finder
To efficiently navigate our portfolio you can use our online product finder by filling in your application and/or product needs and directly order your sample product for delivery.

Formulation tool
Identify the most optimal formulation for your Active Pharmaceutical Ingredient (API) with a simple step by step process or an advanced expert mode.



Global availability

Short replenishment times and local stocks



- DFE Pharma head office Goch
- DFE Pharma salesoffices
- DFE Pharma manufacturing facilities
- DFE Pharma agents and distributors

Notes

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